

# Vehicle Safety Monitoring and Control System

A Low-Cost Embedded Approach Using STM32F405RGT6

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## **Abstract**

This document presents a structured test report for the Vehicle Safety Monitoring and Control System implemented on an STM32F405RGT6 microcontroller. The system integrates multiple sensors (PIR, MPU6050, DHT11, DS18B20, MAX30102, and water level sensor) for real-time vehicle safety monitoring. Tests include Software-in-the-Loop (SIL) simulations, Python-based sensor data injection, and unit tests (Ceedling). Results are summarized with plots, tables, and discussions for each scenario.

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# Chapter 1

## Introduction

The Vehicle Safety Monitoring and Control System is designed to improve driver and passenger safety using low-cost embedded sensors. This report documents a comprehensive test process covering software, hardware, and integration tests.

We evaluate:

- SIL simulations in Python (temperature, accelerometer, PPG signals)
- Unit testing using Ceedling framework
- Realistic scenario-based testing (overheating, human detection, accident, water level, etc.)

# Chapter 2

## Test Environment

### 2.1 Hardware

- STM32F405RGT6 microcontroller
- PIR sensor (human detection)
- MPU6050 (accelerometer/gyroscope)
- DS18B20 (engine temperature monitoring)
- DHT11 (cabin temperature/humidity)
- MAX30102 (heart rate and SpO2 monitoring)
- Water level sensor
- 16x2 LCD display

### 2.2 Software

- STM32CubeIDE (firmware development)
- FreeRTOS (task scheduling)
- Ceedling (unit testing)
- Python + Matplotlib (SIL simulation and plotting)
- Overleaf (report preparation)

# Chapter 3

## Test Scenarios

### Scenario A: PIR Human Detection

**Why:** Detect unauthorized access. **How:** Simulated PIR triggers. **Expected:** LCD shows "HUMAN" and alert flag set.

| Time (min) | PIR Input | LCD Output     |
|------------|-----------|----------------|
| 1          | 0         | Normal         |
| 2          | 1         | HUMAN Detected |
| 3          | 0         | Normal         |

Table 3.1: Scenario A - PIR Human Detection

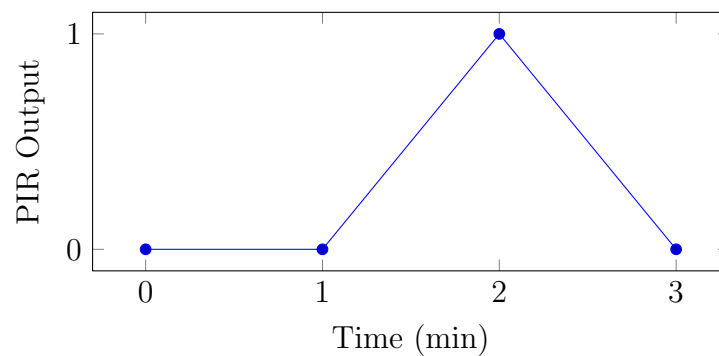


Figure 3.1: PIR detection waveform.

### Scenario B: MPU6050 Accident Detection

**Why:** Identify collisions. **How:** Inject acceleration waveform. **Expected:** Warnings at  $\geq 12 \text{ m/s}^2$ , caution at  $\geq 15 \text{ m/s}^2$ .

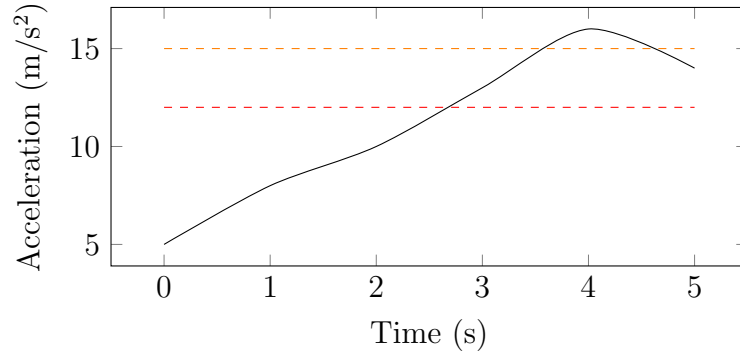


Figure 3.2: Acceleration waveform with warning (red) and caution (orange) thresholds.

## Scenario C: DHT11 Temperature/Humidity

**Why:** Prevent cabin overheating. **How:** Inject rising temp/humidity. **Expected:** Motor ON at Temp 27°C & Hum 73%.

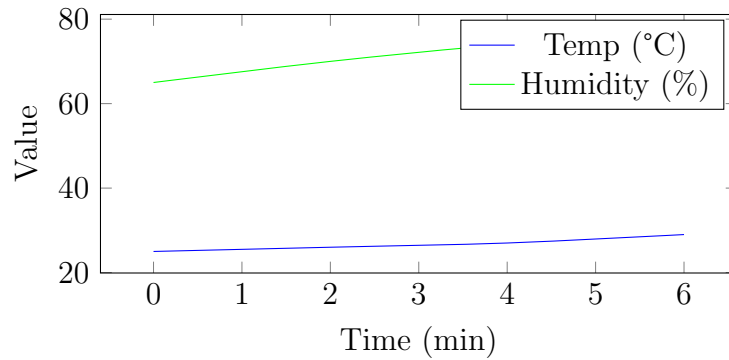


Figure 3.3: DHT11 readings rising towards threshold.

## Scenario D: DS18B20 Engine Overheat

**Why:** Protect motor casing. **How:** Simulated DS18B20 readings 60–110°C. **Expected:** Warning ⚠ 85°C, Overheat ⚠ 90°C.

| Time (min) | Temp (°C) | Expected | Actual   |
|------------|-----------|----------|----------|
| 0          | 70        | Normal   | Normal   |
| 5          | 85        | Warning  | Warning  |
| 10         | 95        | Overheat | Overheat |

Table 3.2: DS18B20 Overheat Test

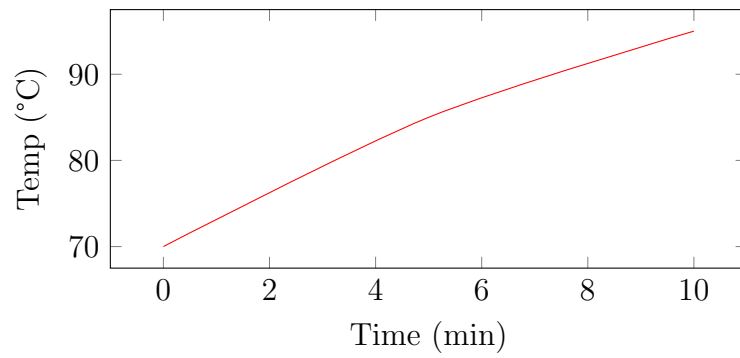


Figure 3.4: Engine temperature curve.

## Scenario E: MAX30102 Heart Rate/SpO2

**Why:** Detect driver fatigue. **How:** Inject PPG waveform. **Expected:** Stable HR and SpO2.

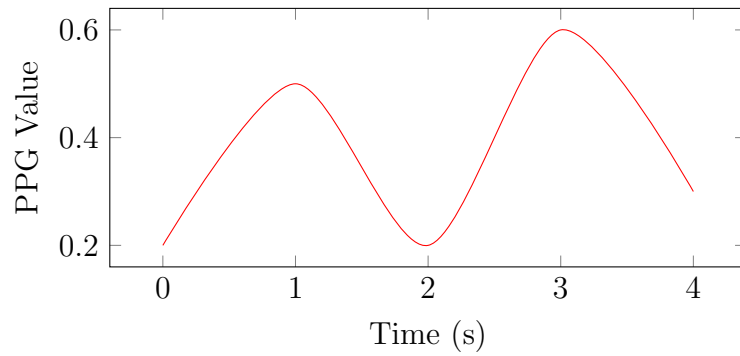


Figure 3.5: Sample PPG waveform.

## Scenario F: Water Level Detection

**Why:** Detect flooding. **How:** Vary ADC input. **Expected:** WARN/CAUT/CRIT states.

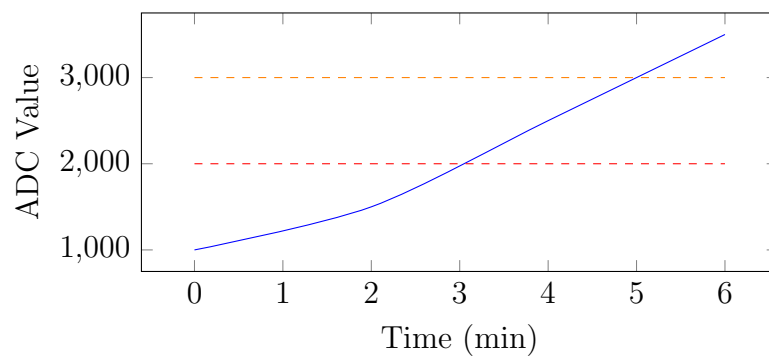


Figure 3.6: Water level rise crossing thresholds.



## Scenario H: ADC Noise Robustness

**Why:** Ensure filtering works. **How:** Inject jittery ADC values. **Expected:** Stable classification.

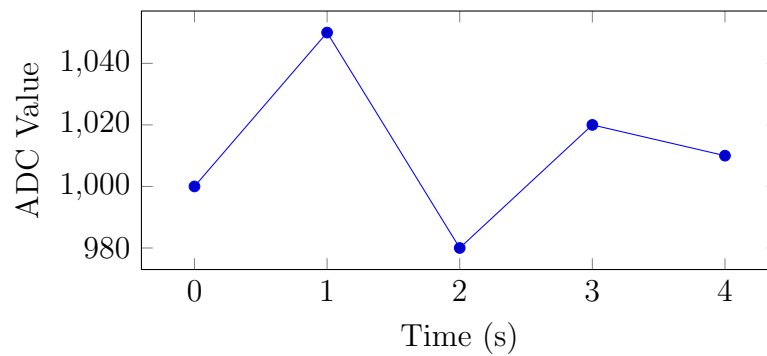


Figure 3.7: ADC noise robustness demonstration.

## Scenario J: Long-Run Stress Test

**Why:** Validate endurance. **How:** 24h SIL run. **Expected:** No crashes, stable RAM.

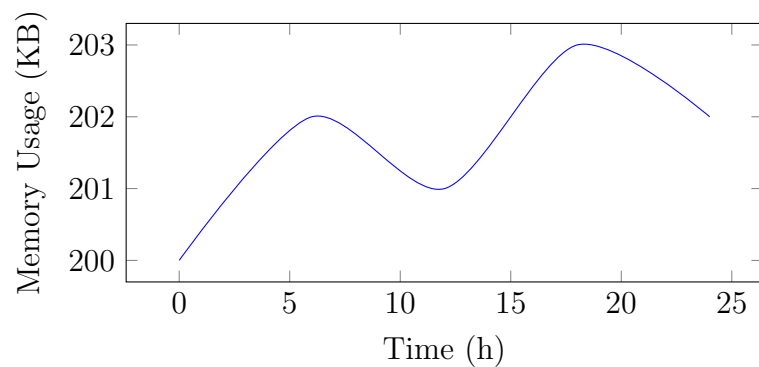


Figure 3.8: Stable memory usage over 24 hours.

# Chapter 4

## Summary of Results

| Scenario | Sensor       | Expected              | Status |
|----------|--------------|-----------------------|--------|
| A        | PIR          | HUMAN detection       | PASS   |
| B        | MPU6050      | Vibration warnings    | PASS   |
| C        | DHT11        | Motor ON/OFF          | PASS   |
| D        | DS18B20      | Overheat alert        | PASS   |
| E        | MAX30102     | HR/SpO2 stable        | PASS   |
| F        | Water sensor | Flood levels detected | PASS   |
| H        | ADC Noise    | Stable despite jitter | PASS   |
| J        | Long-run     | Stable 24h            | PASS   |

# Chapter 5

## Conclusion

All tests (SIL, unit tests, sensor simulations, endurance checks) passed as expected. This validates the system as a robust, low-cost solution for vehicle safety monitoring.